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In [6]: '''
Assignment: 01
Topic: Simple Python Programming
Date: 10/08/2026

1. Write a Python expression to calculate the sum, difference, product,
and quotient of TWO number
'''

a = int(input("Enter first number: "))
b = int(input("Enter 2nd number"))

print("sum, difference, product, and quotient: ", a + b, a - b, a * b, a / b)

Enter first number: 4
Enter 2nd number: 2
sum, difference, product, and quotient: 6 2 8 2.0
```

```
In [7]: #2. Find the remainder when one number is divided by another.

a = int(input("Enter the 1st number: "))
b = int(input("Enter the 2nd number: "))
print("The remainder is: ", a % b)

Enter the 1st number: 5
Enter the 2nd number: 2
The remainder is: 1
```

```
In [8]: #3. Compute the square and cube of a given number.

c = int(input("Enter a number"))
print("The Square and Cube is: ", c**2, c**3)

Enter a number: 2
The Square and Cube is: 4 8
```

```
In [10]: #4. Calculate the average of three numbers using operators.

a = int(input("Enter 1st number:"))
b = int(input("Enter 2nd number:"))
c = int(input("Enter 3rd number:"))
print("the average is:", (a + b + c) / 3)

Enter 1st number: 1
Enter 2nd number: 2
Enter 3rd number: 3
the average is: 2.0
```

```
In [12]: #5. Convert total seconds into minutes and seconds using // and %.

sec = int(input("Enter the second: "))
print("The min is:", sec // 60)

Enter the second: 3700
The min is: 61
```

```
In [1]: #6. Find the area of a circle given the radius.  
r = int(input("Enter the radius:"))  
print("the area of circle is:", 3.14 * r**2)
```

Enter the radius: 7

the area of circle is: 153.86

```
In [ ]: '''  
7. Calculate simple interest using the formula  
SI = (P × R × T) / 100  
'''  
p = int(input("Enter the principle:"))  
r = int(input("Enter the rate:"))  
t = int(input("Enter the time:"))  
print("the simple interest:", (p * r * t) / 100)
```

Enter the principle:2

Enter the rate:3

Enter the time:6

the simple interest: 0.36

```
In [1]: #8. Compute the power of a number using the exponent operator.  
base = int(input("Enter the base value:"))  
exponent = int(input("Enter the exponent: "))  
print("The result= ", base**exponent)
```

Enter the base value: 2

Enter the exponent: 3

The result= 8

```
In [1]: #9. Find the Last digit of a given integer.  
num = int(input("Enter a number: "))  
print("The Last digit of the number is:", num % 10)
```

Enter a number: 1234

The Last digit of the number is: 4

In [2]: *#10. Calculate the total and percentage of 5 subject marks.*

```
sub1 = int(input("Enter subject-1 mark: "))
sub2 = int(input("Enter subject-2 mark: "))
sub3 = int(input("Enter subject-3 mark: "))
sub4 = int(input("Enter subject-4 mark: "))
sub5 = int(input("Enter subject-5 mark: "))
total = sub1+sub2+sub3+sub4+sub5
print("The total mark is:", total)
print("the % of total mark is: ", (total / 500) * 100)
```

```
Enter subject-1 mark: 87
Enter subject-2 mark: 76
Enter subject-3 mark: 87
Enter subject-4 mark: 78
Enter subject-5 mark: 98
```

```
The total mark is: 426
the % of total mark is: 85.2
```

In [1]: *#11. Write a program to demonstrate the use of += and *= operators.*

```
a = 10
print("The intial value is: ", a)
a += 5
print("The Updated value is: ", a)
a *= 2
print("The Updated value is: ", a)
```

```
The intial value is: 10
The Updated value is: 15
The Updated value is: 30
```

In [5]: *#12. Update a variable by adding 10 and then multiplying by 2.*

```
a = int(input("Enter a number: "))
a += 10
a *= 2
print("After Adding and Multiplying a is: ",a)
```

```
Enter a number: 5
```

```
Adding and Multiplying -> 30
```

In [1]: *#13. Write an expression using //= to reduce a number.*

```
num = int(input("Enter a number: "))
num //= 4
print("The reduced number is: ",num)
```

```
Enter a number: 25
The reduced number is: 6
```

```
In [8]: '''  
#14.Use assignment operators to swap two numbers  
without using a temporary variable.  
'''  
  
a = int(input("Enter 1st number:"))  
b = int(input("Enter 2nd number:"))  
a += b  
b = a - b  
a -= b  
print("After swaping:")  
print("a =", a)  
print("b =", b)
```

Enter 1st number: 14
Enter 2nd number: 5

After swaping:
a = 5
b = 14

```
In [9]: #15.Modify a variable value using %= operator.  
var = int(input("Enter a number: "))  
modifier = int(input("Enter modifier: "))  
var %= modifier  
print("After %=", var)
```

Enter a number: 17
Enter modifier: 5

After %= 2

```
In [10]: #16.Find the roots of a quadratic equation.  
a = int(input("Enter a:"))  
b = int(input("Enter b:"))  
c = int(input("Enter c:"))  
d = b**2 - 4*a*c  
root1 = (-b + d**0.5) / 2*a  
root2 = (-b - d**0.5) / 2*a  
print("Root1=", root1)  
print("Root2=", root2)
```

Enter a: 1
Enter b: -5
Enter c: 6

Root1= 3.0
Root2= 2.0

```
In [14]: '''
#17.Accept basic salary from user. Calculate DA(50%), HRA(20%)
and find the gross salary.
'''

basicSalary = float(input("Enter the basic salary: "))
print("DA =",basicSalary * 50 / 100)
print("HRA =",basicSalary * 20 / 100)
print("Gross Salary =", basicSalary + (basicSalary * 50 / 100) + (basicSalary *
```

Enter the basic salary: 20000

DA = 10000.0

HRA = 4000.0

Gross Salary = 34000.0

```
In [15]: #18.Acept temperature in farenhite and convert it into celcius and vice versa.
fa = float(input("Enter temperature in Fahrenheit: "))
print("Temperature in Celsius =", (fa - 32) * 5 / 9)
cel = float(input("\nEnter temperature in Celsius: "))
print("Temperature in Fahrenheit =", (cel * 9 / 5) + 32)
```

Enter temperature in Fahrenheit: 98

Temperature in Celsius = 36.666666666666664

Enter temperature in Celsius: 37

Temperature in Fahrenheit = 98.6

```
In [3]: #19.Find the sum of n natural numbers. "n" must be inputted by the user.
n = int(input("Enter a number: "))
print("sum of n natural numbers is:", n*(n + 1) / 2)
```

Enter a number: 6

sum of n natural numbers is: 21.0

```
In [17]: '''
#20.Accept two numbers and perform the all logical,
bitwise operation and display the result.
'''

a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

print("\n--- Logical Operations ---")
print("a and b =", a and b)
print("a or b  =", a or b)
print("not a   =", not a)
print("not b   =", not b)

print("\n--- Bitwise Operations ---")
print("a & b  =", a & b)
print("a | b  =", a | b)
print("a ^ b  =", a ^ b)
print("~a    =", ~a)
print("~b    =", ~b)
print("a << 1 =", a << 1)
print("b << 1 =", b << 1)
print("a >> 1 =", a >> 1)
print("b >> 1 =", b >> 1)
```

```
Enter first number: 5
Enter second number: 6
```

```
--- Logical Operations ---
a and b = 6
a or b  = 5
not a   = False
not b   = False
```

```
--- Bitwise Operations ---
a & b  = 4
a | b  = 7
a ^ b  = 3
~a     = -6
~b     = -7
a << 1 = 10
b << 1 = 12
a >> 1 = 2
b >> 1 = 3
```

```
In [3]: #21. Write a python program to evaluate the expression  $4x^4 + 3y^3 - 9z + 67$ .
x = int(input("Enter the 1st variable: "))
y = int(input("Enter the 2nd variable: "))
z = int(input("Enter the 3rd variable: "))
print("The result is: ", (4*x**4) + (3*y**3) - 9*z + 67)
```

```
Enter the 1st variable: 4
Enter the 2nd variable: 5
Enter the 3rd variable: 6
The result is: 1412
```

```
In [10]: '''
22. Write a python program to take a input in uppercase and
change it to lower case.
'''

st = ord("A") + 32
print("The lower-case is:",chr(st))

sT = ord("a") - 32
print("The uper-case is",chr(sT))
```

The lower-case is: a
The uper-case is A

```
In [15]: '''
23. Ask the user to enter a number x. Use the sep optional argument to print
out x, 2x, 3x, 4x, and 5x, each separated by three dashes, like below.
Enter a number: 7
7---14---21---28---35
'''

x = int(input("Enter a number: "))
print(x,2*x,3*x,4*x,5*x,sep="---")
```

Enter a number: 7
7---14---21---28---35

```
In [1]: '''
24. Input 3 digit number and calculate the sum of the
digits using // and % operator.
'''

n = int(input("Enter a 3 digit number:"))
hundreds = n // 100
tens = (n // 10) % 10
ones = n % 10
print("The sum of the digit is:", hundreds+tens+ones)
```

Enter a 3 digit number: 123
The sum of the digit is: 6

```
In [2]: #25. Input 3 digit number and reverse the number using arithmetic operators.
n = int(input("Enter a 3 digit number:"))
a = n // 100
b = (n // 10) % 10
c = n % 10
print("The Reverse number is:", c * 100 + b * 10 + a)
```

Enter a 3 digit number: 123
The Reverse number is: 321

```
In [1]: #26. Input two digit number and exchange its digits using operators.
n = int(input("Enter a 2-digit number: "))
tens = n // 10
ones = n % 10
print("After Exchange The digit: ", ones * 10 + tens)
```

Enter a 2-digit number: 23

After Exchange The digit: 32

```
In [3]: '''
27. Input the number of days. Convert it into weeks and
remaining days using // and % operator.
'''
days = int(input("Enter number of days: "))
weeks = days // 7
remaining_days = days % 7
print("Weeks =", weeks)
print("Remaining days =", remaining_days)
```

Enter number of days: 23

Weeks = 3

Remaining days = 2


```
In [4]: '''
#28. Input an amount in rupees and calculate
the number of $500, $200, $100 and $50 notes and
remaining rupees using arithmetic operator.
'''

amount = int(input("Enter amount in rupees: "))

n500 = amount // 500
remaining = amount % 500

n200 = remaining // 200
remaining = remaining % 200

n100 = remaining // 100
remaining = remaining % 100

n50 = remaining // 50
remaining = remaining % 50

print("500 notes =", n500)
print("200 notes =", n200)
print("100 notes =", n100)
print("50 notes =", n50)
print("Remaining rupees =", remaining)
```

Enter amount in rupees: 1850

500 notes = 3
200 notes = 1
100 notes = 1
50 notes = 1
Remaining rupees = 0

```
In [7]: '''
30. Input a 3 digit number and find the 100th digit,
10th digit and unit place digit.
'''

n = int(input("Enter a 3-digit number: "))
hundred = n // 100
ten = (n // 10) % 10
unit = n % 10
print("100th digit =", hundred, "10th digit =", ten, "Unit digit =", unit)
```

Enter a 3-digit number: 257

100th digit = 2 10th digit = 5 Unit digit = 7

In []: